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Wiring board

Abstract

A wiring board includes a pad configured to make an external electrical connection, and an insulating layer. A portion of a lower surface of the pad is covered with the insulating layer. The pad includes a base portion, and an extending portion formed integrally with the base portion and extending toward an outer periphery of a side surface of the base portion in a plan view at a lower end of the side surface of the base portion. The insulating layer is provided with a groove that is located in a periphery of the pad in the plan view, exposes a side surface of the pad, and opens to an upper surface of the insulating layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims priority to Japanese Patent Application No. 2021-171148, filed on Oct. 19, 2021, the entire contents of which are incorporated herein by reference.
FIELD

(2) Certain aspects of the embodiments discussed herein are related to wiring boards, and methods for manufacturing wiring boards.

BACKGROUND

(3) A wiring board includes a pad exposed from an insulating layer disposed at an outermost layer, for example. The pad is used as an external connection pad for making an electrical connection

with a mother board or the like, for example. A method for manufacturing this wiring board includes the steps of forming the external connection pad on an upper surface of a support body, forming an insulating layer on the support body so as to cover the external connection pad, and removing the support body.

(4) An example of the wiring board is proposed in a Japanese Laid-Open Patent Publication No. 2008-141070, for example.

(5) In a case where an area of a top surface of the pad is large, an electrostatic capacitance may become large and deteriorate an electrical signal. For this reason, in recent years, there are demands to reduce the area of the top surface of the pad, in order to reduce the electrostatic capacitance. On the other hand, from a viewpoint of ensuring compatibility with conventional wiring boards with regard to sockets or the like used for making connections to the wiring boards, there are demands not to change a size of an opening in the insulating layer in a periphery of the pad, even when the size of pad is reduced.

(6) In order to satisfy these demands, an outer edge of the pad needs to be moved to the inner side of the outer edge of the opening in the insulating layer in the periphery of the pad in a plan view, without moving the outer edge of the opening in the insulating layer. However, it is difficult to move only the outer edge of the pad to the inner side of the outer edge of the opening in the insulating layer.

SUMMARY

(7) Accordingly, it is an object in one aspect of the embodiments to provide a wiring board in which an outer edge of an external connection pad is disposed on the inner side of an outer edge of an opening in an insulating layer, and a method for manufacturing such a wiring board.

(8) According to one aspect of the embodiments, a wiring includes a pad configured to make an external electrical connection; and an insulating layer, wherein a portion of a lower surface of the pad is covered with the insulating layer, the pad includes a base portion, and an extending portion formed integrally with the base portion and extending toward an outer periphery of a side surface of the base portion in a plan view at a lower end of the side surface of the base portion, and the insulating layer is provided with a groove that is located in a periphery of the pad in the plan view, exposes a side surface of the pad, and opens to an upper surface of the insulating layer.

(9) The object and advantages of the embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

(10) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A and FIG. 1B are diagrams illustrating an example of a wiring board according to a first embodiment;

(2) FIG. 2A, FIG. 2B, and FIG. 2C are diagrams (part 1) illustrating an example of processes of manufacturing the wiring board according to the first embodiment;

(3) FIG. 3A, FIG. 3B, and FIG. 3C are diagrams (part 2) illustrating the example of the processes of manufacturing the wiring board according to the first embodiment;

(4) FIG. 4A, FIG. 4B, and FIG. 4C are diagrams (part 3) illustrating the example of the processes of manufacturing the wiring board according to the first embodiment;

(5) FIG. 5A, FIG. 5B, and FIG. 5C are diagrams (part 4) illustrating the example of the processes of manufacturing the wiring board according to the first embodiment;

(6) FIG. 6A and FIG. 6B are diagrams illustrating an example of the wiring board according to a first modification of the first embodiment;

(7) FIG. 7A and FIG. 7B are diagrams illustrating an example of the wiring board according to a second embodiment;

(8) FIG. 8A and FIG. 8B are diagrams (part 1) illustrating an example of the processes of manufacturing the wiring board according to the second embodiment;

(9) FIG. 9A, FIG. 9B, and FIG. 9C are diagrams (part 2) illustrating the example of the processes of manufacturing the wiring board according to the second embodiment;

(10) FIG. 10A and FIG. 10B are diagrams (part 3) illustrating the example of the processes of manufacturing the wiring board according to the second embodiment;

(11) FIG. 11A and FIG. 11B are diagrams (part 4) illustrating the example of the processes of manufacturing the wiring board according to the second embodiment;

(12) FIG. 12A and FIG. 12B are diagrams (part 5) illustrating the example of the processes of manufacturing the wiring board according to the second embodiment; and

(13) FIG. 13A and FIG. 13B are diagrams illustrating an example of the wiring board according to a first modification of the second embodiment.

DESCRIPTION OF EMBODIMENTS

(14) Preferred embodiments of the present invention will be described with reference to the accompanying drawings. In the drawings, constituent elements that are the same are designated by the same reference numerals, and a repeated description of the same constituent elements may be omitted.

(15) A description will now be given of a wiring board and a method for manufacturing the wiring board according to embodiments and modifications of the present invention.

First Embodiment

(16) [Structure of Wiring Board]

(17) FIG. 1A and FIG. 1B are diagrams illustrating an example of the wiring board according to a first embodiment. FIG. 1A is a partial plan view of the wiring board, and FIG. 1B is a partial cross sectional view of the wiring board along a line A-A in FIG. 1A.

(18) As illustrated in FIG. 1A and FIG. 1B, a wiring board **1** includes a pad **10**, an insulating layer **20**, an interconnect layer **30**, an insulating layer **40**, and an interconnect layer **50**. The wiring board **1** may further include additional insulating layers and interconnect layers.

(19) In the present embodiment, for the sake of convenience, the side of the wiring board **1** provided with the insulating layer **20** in FIG. 1B will be referred to as an upper side or one side, and the side of the wiring board **1** provided with the insulating layer **40** in FIG. 1B will be referred to as a lower side or the other side. In addition, the surface of a portion of the wiring board **1** disposed on the side of the insulating layer **20** will be referred to as an upper surface or one surface, and the surface of the portion of the wiring board **1** disposed on the side of the insulating layer **40** will be referred to as a lower surface or the other surface. However, the wiring board **1** may be used in an up-side-down state, or may be used in a inclined state inclined at an arbitrary angle from a horizontal state, for example. Moreover, a plan view of a target object refers to a view from above the target object in a normal direction to an upper surface **20a** of the insulating layer **20**, and a planar shape of the target object refers to a shape of the target object in the plan view viewed from above the target object in the normal direction to the upper surface **20a** of the insulating layer **20**.

(20) The pad **10** is an external connection pad configured to make an external electrical connection. The pad **10** can be used for making the external electrical connection with a mounting board (not illustrated), such as a mother board or the like, for example. The pad **10** has a laminated structure including a metal layer **11**, and a metal layer **12**.

(21) The metal layer **11** includes a base portion **11b**, and an extending portion **11s**. The extending portion **11s** is formed integrally with the base portion **11b**, and extends on an outer periphery of a side surface of the base portion **11b** in the plan view, at a lower end of the side surface of the base portion **11b**. The extending portion **11s** is formed in an annular shape (or ring shape) on the outer periphery of the side surface of the base portion **11b** in the plan view, for example.

(22) A planar shape of the base portion **11b** may be a circular shape having a diameter in a range of approximately 600 μm to approximately 800 μm , for example. However, the planar shape of the base portion **11b** may be an oval shape, a rectangular shape, or other arbitrary shapes. A width of the extending portion **11s** may be in a range of approximately 10 μm to approximately 30 μm , for example. A thickness of the base portion **11b** may be in a range of approximately 10 μm to approximately 30 μm , for example. A thickness of the extending portion **11s** may be in a range of approximately 0.5 μm to approximately 3.0 μm , for example. An area of a lower surface of the metal layer **11** may be greater than an area of an upper surface of the metal layer **11**. The metal layer **11** may be a copper (Cu) layer, for example.

(23) The metal layer **12** covers a surface of the metal layer **11** exposed from the insulating layer **20**. More particularly, the metal layer **12** continuously covers an upper surface and the side surface of the base portion **11b**, and an upper surface of the extending portion **11s**. The metal layer **12** may be a gold (Au) layer, for example. A thickness of the metal layer **12** may be in a range of approximately 5 μm to approximately 10 μm , for example.

(24) The metal layer **12** may have a laminated structure including a Au layer as an uppermost layer of the laminated structure. The metal layer **12** may be a Ni/Au layer, having a Ni layer on the metal layer **11**, and a Au layer on the Ni layer in this laminating order, a Ni/Pd/Au layer, having a Ni layer on the metal layer **11**, a Pd layer on the Ni layer, and a Au layer on the Pd layer in this laminating order, or the like.

(25) The pad **10** is exposed at the upper surface **20a** of the insulating layer **20**. A portion (excluding a portion connected to a via interconnect) of a lower surface (lower surfaces of the base portion **11b** and the extending portion **11s** of the metal layer **11**) of the pad **10** is covered with the insulating layer **20**. In addition, a side surface of the extending portion **11s** of the metal layer **11** is covered with the insulating layer **20**. A material used for the insulating layer **20** may be an insulating resin or the like having an epoxy-based resin as a main component thereof, for example. The insulating layer **20** may include a filler, such as silica (SiO_2) or the like. A thickness of the insulating layer **20** may be in a range of approximately 10 μm to approximately 70 μm , for example.

(26) An upper surface of the base portion **11b** of the metal layer **11** is located at a position lower than the upper surface **20a** of the insulating layer **20**. A height of the upper surface **20a** of the insulating layer **20**, with reference to the upper surface of the base portion **11b** of the metal layer **11**, may be in a range of approximately 2 μm to approximately 3 μm , for example. An upper surface **10a** of the pad **10** (upper surface of the metal layer **12**) is located at a position higher than the upper surface **20a** of the insulating layer **20**. A height of the upper surface **10a** of the pad **10** (upper surface of the metal layer **12**), with reference to the upper surface **20a** of the insulating layer **20**, may be in a range of approximately 2 μm to approximately 8 μm , for example.

(27) The insulating layer **20** is disposed in a periphery of the pad **10** in the plan view, and exposes a side surface of the pad **10**. A groove **20g**, opening at the upper surface **20a**, is provided in the insulating layer **20**. In a case where the planar shape of the pad **10** is the circular shape, the groove **20g** is formed to a ring shape in the plan view, for example, so that an inner edge and an outer edge of the groove **20g** form circular shapes having different diameters. A width of the groove **20g** may be in a range of approximately 80 μm to approximately 100 μm , for example.

(28) The metal layer **12** covering the side surface of the base portion **11b** of the metal layer **11**, and the metal layer **12** covering the upper surface of the extending portion **11s** of the metal layer **11**, are exposed from the insulating layer **20** inside the groove **20g**. Among surfaces formed by the insulating layer **20** and exposed inside the groove **20g**, when the surface at a lowest position is defined as a bottom surface of the groove **20g**, the upper surface of the extending portion **11s** is located at a position lower than the bottom surface of the groove **20g**. A height of the bottom surface of the groove **20g**, with reference to the upper surface of the extending portion **11s**, may be in a range of approximately 2 μm to approximately 3 μm , for example. In addition, in the metal layer **12**, when an upper surface of the portion laminated on the extending portion **11s** is defined as

an outer peripheral surface of the metal layer **12**, the outer peripheral surface of the metal layer **12** is located at a position higher than the bottom surface of the groove **20g**. A height of the outer peripheral surface of the metal layer **12**, with reference to the bottom surface of the groove **20g**, may be in a range of approximately 2 μm to approximately 8 μm , for example.

(29) The interconnect layer **30** is formed on the other side of the insulating layer **20**. The interconnect layer **30** includes a via interconnect filling a via hole **20x** that penetrates the insulating layer **20** and exposes the lower surface of the pad **10**, a via receiving pad and an interconnect pattern formed on a lower surface of the insulating layer **20**, for example. The via interconnect penetrates the insulating layer **20**, and makes contact with the lower surface of the pad **10**. The via hole **20x** may be a cavity having a truncated cone shape, and a diameter of an opening of the cavity at one end which opens to the insulating layer **40** is greater than a diameter of a bottom surface of the cavity formed by the lower surface of the pad **10**. A material used for the interconnect layer **30** may be copper or the like, for example. A thickness of the via receiving pad and the interconnect pattern forming the interconnect layer **30** may be in a range of approximately 10 μm to approximately 30 μm , for example.

(30) The insulating layer **40** is formed on the lower surface of the insulating layer **20**, so as to cover the interconnect layer **30**. A material used for the insulating layer **40** and a thickness of the insulating layer **40** may be the same as those of the insulating layer **20**, for example. The insulating layer **40** may include a filler, such as silica (SiO_2) or the like.

(31) The interconnect layer **50** is formed on the other side of the insulating layer **40**. The interconnect layer **50** includes a via interconnects filling a via hole **40x** that penetrates the insulating layer **40** and exposes a lower surface of the via receiving pad of the interconnect layer **30**, a pad and an interconnect pattern formed on a lower surface of the insulating layer **40**, for example. The via hole **40x** may be a cavity having a truncated cone shape, and a diameter of an opening of the cavity at one end which opens to the lower surface the insulating layer **40** is greater than a diameter of a bottom surface of the cavity formed by the lower surface of the via receiving pad. A material used for the interconnect layer **50**, and a thickness of the via receiving pad and the interconnect pattern forming the interconnect layer **50**, may be the same as those of the interconnect layer **30**, for example.

(32) As described above, in the wiring board **1**, the insulating layer **20** is provided with the groove **20g**, that is located in the periphery of the pad **10** in the plan view, and opens to the upper surface **20a** of the insulating layer **20**. For this reason, an outer edge of the pad **10** can be located on the inner side of an outer edge of the opening (an outer edge of the groove **20g**) of the insulating layer **20** in the plan view, thereby enabling an area of the upper surface **10a** of the pad **10** to be reduced. By reducing the area of the upper surface **10a** of the pad **10**, it becomes possible to reduce an electrostatic capacitance of the pad **10**, and to reduce a deterioration of an electrical signal passing through the pad **10**.

(33) If the groove **20g** were not provided, a position of the opening in the insulating layer **20**, exposing the upper surface **10a** of the pad **10**, would be the position where an edge defining the opening coincides with the outer edge of the pad **10** in the plan view. In this case, if the area of the upper surface **10a** of the pad **10** were made smaller than that of the conventional case, the size of the opening would also become smaller, thereby making it impossible to ensure compatibility with the conventional wiring boards.

(34) That is, although a socket is used when the pad **10** is connected to the mounting board, such as the mother board or the like, the opening in the insulating layer **20** that exposes the upper surface **10a** of the pad **10** has a size such that the socket does not make contact with the insulating layer **20**. Accordingly, in order to reduce the electrostatic capacitance of the pad **10** while ensuring the compatibility with the conventional wiring boards, it is necessary to reduce the area of the upper surface **10a** of the pad **10** without changing the size of the opening in the insulating layer **20**. By providing the groove **20g**, that is located in the periphery of the pad **10** in the plan view, and opens

to the upper surface **20a** of the insulating layer **20**, it is possible to reduce the area of the upper surface **10a** of the pad **10** while ensuring compatibility with the conventional circuit boards.

(35) Further, when the pad **10** and the mounting board, such as the mother board or the like, are connected via a solder, excess solder flows from the upper surface **10a** of the pad **10** into the groove **20g**. Hence, it is possible to reduce short-circuiting of mutually adjacent pads. In addition, because the solder flows into the groove **20g**, the upper surface **10a** and the side surface of the pad **10** are bonded three-dimensionally with the solder, thereby improving a bonding strength between the pad **10** and the solder.

(36) Moreover, because a stepped portion is formed at a boundary between the bottom surface of the groove **20g** and the outer peripheral surface of the metal layer **12**, corner portions contacted by the solder flowing into the groove **20g** increases. Accordingly, because a force applied to the solder is dispersed, it is possible to improve a durability of the solder.

(37) [Method for Manufacturing Wiring Board]

(38) Next, a method for manufacturing the wiring board according to the first embodiment will be described. FIG. 2A through FIG. 5C are diagrams illustrating an example of processes of manufacturing the wiring board according to the first embodiment. In the present embodiment, the processes of manufacturing one wiring board is described. However, a plurality of parts respectively corresponding to the wiring boards may be formed simultaneously, and the plurality of parts may thereafter be singulated into individual wiring boards.

(39) First, in the process illustrated in FIG. 2A, a support body **300** having a flat upper surface is prepared. Although a metal plate, a metal foil, or the like can be used for the support body **300**, an example in which a copper foil is used as the support body **300** will be described in the present embodiment. A thickness of the support body **300** may be in a range of approximately 18 μm to approximately 100 μm , for example.

(40) Next, sacrificial layers **310** and **320** are formed on the support body **300**. The sacrificial layer **310** is formed so as to cover an entire upper surface of the support body **300**, for example. The sacrificial layer **320** is formed so as to cover an entire upper surface of the sacrificial layer **310**, for example. The sacrificial layer **320** is formed of a material that can be selectively etched with respect to the metal layer **11** in a subsequent process. As an example, the sacrificial layer **310** is a copper layer, and the sacrificial layer **320** is a nickel layer. A thickness of the sacrificial layer **310** may be in a range of approximately 1 μm to approximately 5 μm , for example. A thickness of the sacrificial layer **320** may be in a range of approximately 1 μm to approximately 5 μm , for example. It is not essential to provide the sacrificial layer **310**, which is the copper layer, on the support body **300**. However, by providing the sacrificial layer **310**, which is the copper layer, on the support body **300**, it is possible to obtain an effect of canceling unevenness of (or irregularities on) the upper surface of the support body **300**, and to obtain an effect of improving adhesion to the support body **300**.

(41) Next, in the process illustrated in FIG. 2B, an annular (ring shaped) sacrificial layer **330**, protruding from an upper surface of the sacrificial layer **320**, is formed on the support body **300**. More particularly, a resist layer **400** (for example, a dry film resist or the like) having an opening **400x** at a portion where the sacrificial layer **330** is to be formed, is first formed at a predetermined position on the upper surface of the sacrificial layer **320**. The opening **400x** is formed to a ring shape so that an inner edge and an outer edge of the opening **400x** form circular shapes having different diameters in the plan view, for example. Then, the sacrificial layer **330** is formed on the upper surface of the sacrificial layer **320** exposed inside the opening **400x** in the resist layer **400**, by an electrolytic plating or the like utilizing the support body **300**, the sacrificial layer **310**, and the sacrificial layer **320** as a plating power supply layer. The sacrificial layer **330** is formed of a material that can be selectively etched with respect to the metal layer **11** in a subsequent process. As an example, the sacrificial layer **330** is a nickel layer. A width of the sacrificial layer **330** may be in a range of approximately 5 μm to approximately 30 μm , for example. A thickness of the

sacrificial layer **330** may be in a range of approximately 5 μm to approximately 20 μm , for example. After forming the sacrificial layer **330**, in the process illustrated in FIG. 2C, the resist layer **400** is removed. The sacrificial layer **320** is a typical example of a first sacrificial layer according to the present invention. In addition, the sacrificial layer **330** is a typical example of a second sacrificial layer according to the present invention.

(42) Next, in the processes illustrated in FIG. 3A and FIG. 3B, the metal layer **11** is formed so as to fill a recess formed by the sacrificial layer **320** and the sacrificial layer **330**, cover the upper surface and an inner periphery of the sacrificial layer **330**, and expose an outer periphery of the sacrificial layer **330**. More particularly, first, in the process illustrated in FIG. 3A, a resist layer **410** (for example, a dry film resist or the like) is formed at a predetermined position on the upper surface of the sacrificial layer **320**, covering the outer periphery of the upper surface of the sacrificial layer **330** having the ring shape in the plan view, and including an opening **410x** exposing the inner periphery of the sacrificial layer **330**. The opening **410x** is formed to a circular shape in the plan view, for example. In the plan view inside the opening **410x**, the inner periphery of the upper surface of the sacrificial layer **330**, and the upper surface of the sacrificial layer **320** surrounded by the sacrificial layer **330**, are exposed.

(43) Next, in the process illustrated in FIG. 3B, the metal layer **11** is formed on the upper surface of the sacrificial layer **320** and the upper surface and the inner periphery of the sacrificial layer **330** that are exposed inside the opening **410x** in the resist layer **410**, by an electrolytic plating or the like utilizing the support body **300**, the sacrificial layer **310**, and the sacrificial layer **320**, and the sacrificial layer **330** as a plating power supply layer. The metal layer **11** includes the base portion **11b** and the extending portion **11s**. A portion of the metal layer **11** formed on the upper surface of the sacrificial layer **330** exposed inside the opening **410x** forms the extending portion **11s**, and other portions of the metal layer **11** form the base portion **11b**. The metal layer **11** is formed of a material different from that of the sacrificial layers **320** and **330**, so that the sacrificial layers **320** and **330** can be selectively etched in a subsequent process. In this example, because the sacrificial layers **310** and **320** are nickel layers, the metal layer **11** is a copper layer.

(44) Next, in the process illustrated in FIG. 3C, the insulating layer **20** that covers the sacrificial layer **320**, the sacrificial layer **330**, and the metal layer **11**, is formed on the support body **300**. More particularly, first, after removing the resist layer **410**, a film of an epoxy-based resin or the like in a semi-cured state is laminated on the support body **300**, and cured to form the insulating layer **20**. The insulating layer **20** is formed so as to cover the upper surface of the sacrificial layer **320** exposed from the metal layer **11**, the upper surface and the outer peripheral side surface of the sacrificial layer **330** exposed from the metal layer **11**, and the upper surface and the side surface of the metal layer **11** exposed from the sacrificial layer **330**. Alternatively, instead of laminating the film of the epoxy-based resin or the like in the semi-cured state, a liquid or paste of the epoxy-based resin or the like may be coated and thereafter cured, so as to form the insulating layer **20**. The thickness or the like of the insulating layer **20** are as described above.

(45) Next, in the process illustrated in FIG. 4A, the via hole **20x** is formed in the insulating layer **20**, so as to penetrate the insulating layer **20** and expose the upper surface of the metal layer **11**. The via hole **20x** can be formed by laser beam machining using a CO.sub.2 laser or the like, for example. Thereafter, a desmear process may be performed to remove residual resin adhered to the upper surface of the metal layer **11** exposed at the bottom of the via hole **20x**.

(46) Next, in the process illustrated in FIG. 4B, the interconnect layer **30** is formed on the insulating layer **20**. The interconnect layer **30** includes the via interconnect filling the via hole **20x**, the via receiving pad formed on the insulating layer **20**, and the interconnect pattern, for example. The interconnect layer **30** is electrically connected to the metal layer **11** exposed at the bottom of the via hole **20x**. A material used for the interconnect layer **30** may be copper (Cu) or the like, for example. The interconnect layer **30** can be formed using various interconnect layer forming methods, such as a semi-additive method, a subtractive method, or the like.

(47) Next, in the process illustrated in FIG. 4C, the same processes as those described above in conjunction with FIG. 3C through FIG. 4B are repeated to form the insulating layer **40** on the interconnect layer **30**, form the via hole **40x** exposing the upper surface of the via receiving pad of the interconnect layer **30** in the insulating layer **40**, and further form the interconnect layer **50**. The material and thickness of the insulating layer **40** may be the same as those of the insulating layer **20**, for example. The material and thickness of the interconnect layer **50** may be the same as those of the interconnect layer **30**, for example.

(48) Next, in the processes illustrated in FIG. 5A through FIG. 5C, the support body **300**, the sacrificial layer **310**, the sacrificial layer **320**, and the sacrificial layer **330** are removed by etching, and the external connection pad **10** including the metal layer **11** is formed. More particularly, first, in the process illustrated in FIG. 5A, the support body **300** and the sacrificial layer **310** illustrated in FIG. 4C are removed. The support body **300**, which is a copper foil, and the sacrificial layer **310**, which is a copper layer, can be removed by wet etching using a hydrogen peroxide/sulfuric acid-based aqueous solution, a sodium persulfate aqueous solution, an ammonium persulfate aqueous solution, or the like, for example. The sacrificial layer **320**, which is a nickel layer, serves as an etching stopper layer, so that only the support body **300** and the sacrificial layer **310** are removed by the wet etching. FIG. 5A illustrates the structure in an up-side-down state with respect to the structure illustrated in FIG. 4C or the like. The same applies to FIG. 5B and FIG. 5C that will be described later.

(49) Next, in the process illustrated in FIG. 5B, the sacrificial layers **320** and **330** illustrated in FIG. 5A are removed. The sacrificial layers **320** and **330**, that are nickel layers, can be removed by wet etching using a hydrogen peroxide/nitric acid-based aqueous solution or the like, for example. Because copper is not removed by the etchant for removing the nickel layer, the metal layer **11** will not be etched by this etchant. As a result, the groove **20g** is formed in the insulating layer **20**, at the position in the periphery of the metal layer **11** in the plan view, and opening to the upper surface **20a** of the insulating layer **20**. For example, the groove **20g** is formed in the ring shape in the plan view, so that the inner edge and the outer edge of the groove **20g** form the circular shapes having the different diameters. In addition, in the metal layer **11**, the side surface of the base portion **11b**, and the upper surface of the extending portion **11s**, are exposed inside the groove **20g**. In this state, the upper surface of the base portion **11b** approximately coincides with the upper surface **20a** of the insulating layer **20**. The upper surface of the extending portion **11s** approximately coincides with the bottom surface of the groove **20g** of the insulating layer **20**.

(50) Next, if necessary in order to remove the oxide film famed on the upper surface and the side surface of the base portion **11b** and the upper surface of the extending portion **11s**, the upper surface and the side surface of the base portion **11b** and the upper surface of the extending portion **11s** are etched by an amount in a range of approximately 2 μm to approximately 3 μm , for example. Thus, as illustrated in FIG. 5B, the upper surface of the base portion **11b** becomes exposed at a position recessed by an amount in a range of approximately 2 μm to approximately 3 μm from the upper surface **20a** of the insulating layer **20**. Further, the upper surface of the extending portion **11s** becomes exposed at a position recessed by an amount in a range of approximately 2 μm to approximately 3 μm from the bottom surface of the groove **20g** of the insulating layer **20**.

(51) Next, in the process illustrated in FIG. 5C, the metal layer **12** is formed on the surface of the metal layer **11** exposed from the insulating layer **20**, by electroless plating, for example. As a result, the wiring board **1** is completed. The metal layer **12** may be an Au layer, a Ni/Au layer (a metal layer in which a Ni layer and a Au layer are laminated in this order on the pad **10**), a Ni/Pd/Au layer (a metal layer in which a Ni layer, a Pd layer, and a Au layer are laminated in this order on the pad **10**), or the like, for example.

(52) In this process, the metal layer **12** covering the side surface of the base portion **11b** of the metal layer **11**, and the metal layer **12** covering the upper surface of the extending portion **11s** of the metal layer **11**, are exposed from the insulating layer **20** inside the groove **20g**. In addition, the

upper surface of the metal layer **12** is located at a position higher than the upper surface **20a** of the insulating layer **20** by an amount in a range of approximately 2 μm to approximately 8 μm , for example. Moreover, the outer peripheral surface of the metal layer **12** is located at a position higher than the bottom surface of the groove **20g** by an amount in a range of approximately 2 μm to approximately 8 μm , for example.

(53) <First Modification of First Embodiment>

(54) A first modification of the first embodiment relates to an example of the wiring board having a pad with a different layer structure. In the first modification of the first embodiment, a description of the constituent elements that are the same as those of the embodiment described above may be omitted.

(55) FIG. **6A** and FIG. **6B** are diagrams illustrating an example of the wiring board according to a first modification of the first embodiment. FIG. **6A** is a partial plan view of the wiring board, and FIG. **6B** is a partial cross sectional view of the wiring board along a line B-B in FIG. **6A**.

(56) As illustrated in FIG. **6A** and FIG. **6B**, a wiring board **1A** differs from the wiring board **1** illustrated in FIG. **1A**, FIG. **1B**, or the like in that the pad **10** has a single-layer structure. The pad **10** is a copper layer, for example. The pad **10** includes the base portion **11b** and the extending portion **11s**, but does not include the metal layer **12**.

(57) The upper surface **10a** of the pad **10** (the upper surface of the base portion **11b**) is located at a position lower than the upper surface **20a** of the insulating layer **20**. The height of the upper surface **20a** of the insulating layer **20**, with reference to the upper surface of the base portion **11b** of the pad **10**, may be in a range of approximately 2 μm to approximately 3 μm , for example. The upper surface of the extending portion **11s** is located at a position lower than the bottom surface of the groove **20g**. The height of the bottom surface of the groove **20g**, with reference to the upper surface of the extending portion **11s**, may be in a range of approximately 2 μm to approximately 3 μm , for example.

(58) An organic film, that covers the upper surface and the side surface of the base portion **11b** of the pad **10**, and the upper surface of the extending portion **11s** of the pad **10**, may be provided, as required. The organic film includes an azole compound, an imidazole compound, or the like, for example. A thickness of the organic film may be approximately 1 μm or less, for example. Although the illustration of the organic film is omitted in FIG. **6B** to simplify the drawing, a coverage provided by the organic film will be the same as the coverage provided by the metal layer **12** in FIG. **5C**, for example.

(59) The wiring board **1A** illustrated in FIG. **6A** and FIG. **6B** can be manufactured by processes that are the same as those of the first embodiment described above in conjunction with FIG. **2A** through FIG. **5B**. After the process illustrated in FIG. **5B**, the organic film can be formed by performing an Organic Solderability Preservative (OSP) process with respect to the base portion **11b** and the extending portion **11s**, as required.

(60) In a case where the pad **10** has the single-layer structure and the organic film is not formed, the etching process for removing the oxide film is not required.

Second Embodiment

(61) A second embodiment relates to an example of the wiring board having the insulating layer including the groove with a different shape. In the second embodiment, a description of the constituent elements that are the same as those of the embodiment described above may be omitted.

(62) FIG. **7A** and FIG. **7B** are diagrams illustrating the wiring board according to the second embodiment. FIG. **7A** is a partial plan view of the wiring board, and FIG. **7B** is a partial cross sectional view of the wiring board along a line C-C in FIG. **7A**.

(63) As illustrated in FIG. **7A** and FIG. **7B**, a wiring board **2** differs from the wiring board **1** illustrated in FIG. **1A**, FIG. **1B**, or the like in that a stepped surface **20s** is formed on the side surface of the groove **20g** of the insulating layer **20**, and that the upper surface **10a** of the pad **10** is located at a position recessed with respect to the upper surface **20a** of the insulating layer **20**. The

height of the upper surface **20a** of the insulating layer **20**, with reference to the upper surface **10a** of the pad **10**, may be in a range of approximately 5 μm to approximately 20 μm , for example.

(64) In the insulating layer **20**, the stepped surface **20s** is located between the upper surface **20a** of the insulating layer **20** and the bottom surface of the groove **20g**, along a thickness direction of the wiring board **2**. The stepped surface **20s** is approximately parallel to the upper surface **20a** of the insulating layer **20** and the bottom surface of the groove **20g**. The upper surface of the base portion **11b** is located at a position lower than the stepped surface **20s**. A height of the stepped surface **20s**, with reference to the upper surface of the base portion **11b**, may be in a range of approximately 2 μm to approximately 3 μm , for example. In addition, the upper surface **10a** of the pad **10** (the upper surface of the metal layer **12**) is located at a position higher than the stepped surface **20s**. A height of the upper surface **10a** of the pad **10** (the upper surface of the metal layer **12**), with reference to the stepped surface **20s**, may be in a range of approximately 2 μm to approximately 8 μm , for example.

(65) In the case where the planar shape of the pad **10** is the circular shape, the groove **20g** can be formed to the ring shape in the plan view, for example, so that the inner edge and the outer edge of the groove **20g** form the circular shapes having the different diameters. In the groove **20g**, a width of a portion on the lower side of the stepped surface **20s** may be in a range of approximately 80 μm to approximately 100 μm , for example. Moreover, a width of the stepped surface **20s** may be in a range of approximately 10 μm to approximately 30 μm , for example.

(66) The wiring board **2** illustrated in FIG. 7A and FIG. 7B can be formed in the following manner, for example. First, in the processes illustrated in FIG. 8A through FIG. 9B, sacrificial layers **340** and **350** are successively formed in a predetermined region on the support body **300**, and further, an annular sacrificial layer **360** protruding from an upper surface of the sacrificial layer **350** is formed. The sacrificial layer **350** is a typical example of the first sacrificial layer according to the present invention. The sacrificial layer **360** is a typical example of the second sacrificial layer according to the present invention.

(67) More particularly, first, in the process illustrated in FIG. 8A, a resist layer **420** (for example, a dry film resist or the like) having an opening **420x** is formed in a predetermined region on the upper surface of the support body **300**. Then, the sacrificial layers **340** and **350** are successively laminated on the upper surface of the support body **300** exposed inside the opening **420x** in the resist layer **420**, by electrolytic plating or the like utilizing the support body **300** as a plating power supply layer. Thereafter, the resist layer **420** is removed. As an example, the sacrificial layer **340** is a copper layer, and the sacrificial layer **350** is a nickel layer. The sacrificial layers **340** and **350** are metal layers that are ultimately removed by etching. A thickness of the sacrificial layer **340** may be in a range of approximately 5 μm to approximately 20 μm , for example. A thickness of the sacrificial layer **350** may be in a range of approximately 1 μm to approximately 10 μm , for example. It is not essential to provide the sacrificial layer **340**, which is a copper layer, on the support body **300**. However, by providing the sacrificial layer **340**, which is the copper layer, on the support body **300**, it is possible to obtain an effect of canceling unevenness of (or irregularities on) the upper surface of the support body **300**, and to obtain an effect of improving adhesion to the support body **300**.

(68) Next, in the process illustrated in FIG. 8B, a resist layer **430** (for example, a dry film resist or the like), including an opening **430x** that opens an outer periphery of an upper surface of the sacrificial layer **350** in a ring shape, is formed on the upper surface of the support body **300**. In order to prevent a misalignment (or positional error) with the upper surface of the sacrificial layer **350**, the opening **430x** is preferably not provided in an outermost peripheral portion of the upper surface of the sacrificial layer **350**. In other words, the outermost peripheral portion of the upper surface of the sacrificial layer **350** is preferably covered with the resist layer **430**.

(69) Next, in the process illustrated in FIG. 9A, an annular (or ring shaped) sacrificial layer **360**, protruding from the upper surface of the sacrificial layer **350**, is formed on the support body **300**.

More particularly, the sacrificial layer **360** is formed on the upper surface of the sacrificial layer **350** exposed inside the opening **430x** in the resist layer **430**, by electrolytic plating or the like utilizing the support body **300**, the sacrificial layer **350**, and the sacrificial layer **350** as a plating power supply layer. The sacrificial layer **360** is formed of a material that can be selectively etched with respect to the metal layer **11** in a subsequent process. As an example, the sacrificial layer **360** is a nickel layer. A width of the sacrificial layer **360** may be in a range of approximately 5 μm to approximately 30 μm , for example. A thickness of the sacrificial layer **360** may be in a range of approximately 10 μm to approximately 30 μm , for example. After forming the sacrificial layer **360**, in the process illustrated in FIG. **9B**, the resist layer **430** is removed. The sacrificial layer **340** and the sacrificial layer **350** are formed at positions overlapping each other in the plan view, and an outer edge of the sacrificial layer **360** is formed inside the outer edges of the sacrificial layers **340** and **350** in the plan view.

(70) Next, in the processes illustrated in FIG. **9C** and FIG. **10A**, the metal layer **11** is formed so as to fill a recess formed by the sacrificial layer **350** and the sacrificial layer **360**, cover an inner periphery of the upper surface of the sacrificial layer **360**, and expose an outer periphery of the sacrificial layer **360**. More particularly, first, in the process illustrated in FIG. **9C**, a resist layer **440** (for example, a dry film resist or the like) is formed at a predetermined position on the upper surface of the support body **300**, covering the outer periphery of the upper surface of the sacrificial layer **360** having the ring shape in the plan view, and including an opening **440x** exposing the inner periphery of the sacrificial layer **360**. The opening **440x** is formed to a circular shape in the plan view, for example. In the plan view inside the opening **440x**, the inner periphery of the upper surface of the sacrificial layer **360**, and the upper surface of the sacrificial layer **350** surrounded by the sacrificial layer **360**, are exposed.

(71) Next, in the process illustrated in FIG. **10A**, the metal layer **11** is formed on the upper surface of the sacrificial layer **350** exposed inside the opening **440x** in the resist layer **440**, and the upper surface and the inner side surface of the sacrificial layer **360**, by electrolytic plating or the like utilizing the support body **300**, the sacrificial layer **340**, the sacrificial layer **350**, and the sacrificial layer **360** as a plating power supply layer. The metal layer **11** includes the base portion **11b** and the extending portion **11s**. A portion of the metal layer **11** formed on the upper surface of the sacrificial layer **360** exposed inside the opening **440x** forms the extending portion **11s**, and other portions of the metal layer **11** form the base portion **11b**. The metal layer **11** is formed of a material different from that of the sacrificial layers **350** and **360**, so that the sacrificial layers **350** and **360** can be selectively etched in a subsequent process. In this example, because the sacrificial layers **350** and **360** are nickel layers, the metal layer **11** is a copper layer.

(72) Next, in the process illustrated in FIG. **10B**, the insulating layer **20** is formed on the support body **300**, so as to cover the sacrificial layer **340**, the sacrificial layer **350**, the sacrificial layer **360**, and the metal layer **11**. More particularly, first, after removing the resist layer **440**, a film of an epoxy-based resin or the like in a semi-cured state is laminated on the support body **300**, and cured to form the insulating layer **20**. The insulating layer **20** is formed so as to cover the upper surface and the outer side surface of the sacrificial layer **360** exposed from the metal layer **11**, the upper surface and the side surface of the metal layer **11** exposed from the sacrificial layer **360**, the outer periphery of the upper surface and the side surface of the sacrificial layer **350**, and the side surface of the sacrificial layer **340**. Alternatively, instead of laminating the film of the epoxy-based resin or the like in the semi-cured state, a liquid or paste of the epoxy-based resin or the like may be coated and thereafter cured, so as to form the insulating layer **20**. The thickness or the like of the insulating layer **20** are as described above.

(73) Next, in the process illustrated in FIG. **11A**, the same processes as those described above in conjunction with FIG. **4A** through FIG. **4C** are performed to form the via hole **20x** in the insulating layer **20** to expose the upper surface of the metal layer **11**, and to further form the interconnect layer **30**. Then, the insulating layer **40** is formed on the interconnect layer **30**, the via hole **40x** is

formed in the insulating layer **40** to expose the upper surface of the via receiving pad of the interconnect layer **30**, and the interconnect layer **50** is further formed.

(74) Next, in the processes illustrated in FIG. **11B** through FIG. **12B**, the support body **300**, the sacrificial layer **340**, the sacrificial layer **350**, and the sacrificial layer **360** are removed by etching, and the external connection pad **10** including the metal layer **11** is formed. More particularly, first, in the process illustrated in FIG. **11B**, the support body **300** illustrated in FIG. **11A** is removed. The support body **300** can be removed by mechanical stripping, for example. In this case, the sacrificial layer **340** is not removed. The support body **300** made of the copper foil can be removed by wet etching using a hydrogen peroxide/sulfuric acid-based aqueous solution, a sodium persulfate aqueous solution, an ammonium persulfate aqueous solution, or the like, for example. In this case, the sacrificial layer **340**, which is a copper layer, is also removed. FIG. **11B** illustrates the structure in an up-side-down state with respect to the structure illustrated in FIG. **10B** or the like. The same applies to FIG. **12A** and FIG. **12B** that will be described later.

(75) Next, in the process illustrated in FIG. **12A**, the sacrificial layers **340**, **350**, and **360** illustrated in FIG. **11B** are removed. The sacrificial layer **340**, which is a copper layer, can be removed by wet etching using an etchant such as a hydrogen peroxide/sulfuric acid-based aqueous solution, a sodium persulfate aqueous solution, an ammonium persulfate aqueous solution, or the like, for example. The sacrificial layers **350** and **360**, which are nickel layers, can be removed by wet etching using a hydrogen peroxide/nitric acid-based aqueous solution or the like, for example. In a case where the sacrificial layer **340** is removed in the process illustrated in FIG. **11B**, the etching of the sacrificial layer **340** is not required. Because copper is not removed by the etchant for the nickel layer, the metal layer **11** will not be etched by this etchant. As a result, the groove **20g** is formed in the insulating layer **20**, at the position in the periphery of the metal layer **11** in the plan view, and opening to the upper surface **20a** of the insulating layer **20**. For example, the groove **20g** is formed in the ring shape in the plan view, so that the inner edge and the outer edge of the groove **20g** form the circular shapes having the different diameters. In addition, the stepped surface **20s** is formed on the side surface of the groove **20g** of the insulating layer **20**. Moreover, in the metal layer **11**, the upper surface of the base portion **11b** is exposed at a position that is recessed with respect to the upper surface **20a** of the insulating layer **20**, and the side surface of the base portion **11b**, and the upper surface of the extending portion **11s**, are exposed inside the groove **20g**. In this state, the upper surface of the base portion **11b** approximately coincides with the stepped surface **20s** of the insulating layer **20**. The upper surface of the extending portion **11s** approximately coincides with the bottom surface of the groove **20g** of the insulating layer **20**.

(76) Next, if necessary in order to remove the oxide film formed on the upper surface and the side surface of the base portion **11b** and the upper surface of the extending portion **11s**, the upper surface and the side surface of the base portion **11b** and the upper surface of the extending portion **11s** are etched by an amount in a range of approximately 2 μm to approximately 3 μm , for example. Thus, as illustrated in FIG. **12A**, the upper surface of the base portion **11b** becomes exposed at a position recessed by an amount in a range of approximately 2 μm to approximately 3 μm from the stepped surface **20s** of the insulating layer **20**. Further, the upper surface of the extending portion **11s** becomes exposed at a position recessed by an amount in a range of approximately 2 μm to approximately 3 μm from the bottom surface of the groove **20g** of the insulating layer **20**.

(77) Next, in the process illustrated in FIG. **12B**, the metal layer **12** is formed on the surface of the metal layer **11** exposed from the insulating layer **20**, by electroless plating, for example. The metal layer **12** is as described above. As a result, the wiring board **2** is completed.

(78) In this process, the metal layer **12** covering the side surface of the base portion **11b** of the metal layer **11**, and the metal layer **12** covering the upper surface of the extending portion **11s** of the metal layer **11**, are exposed from the insulating layer **20** inside the groove **20g**. The upper surface of the metal layer **12** is located at a position higher than the stepped surface **20s** by an amount in a range of approximately 2 μm to approximately 8 μm , for example. Further, the outer

peripheral surface of the metal layer **12** is located at a position higher than the bottom surface of the groove **20g** by an amount in a range of approximately 2 μm to approximately 8 μm , for example. (79) The wiring board **2** has the following advantageous features in addition to the advantageous features of the wiring board **1**. That is, in the wiring board **2**, because the upper surface **10a** of the pad **10** is located at a position further lower than the upper surface **20a** of the insulating layer **20**, a center of gravity of the entire solder is located closer to a center of the wiring board **2**, when compared to the wiring board **1**. For this reason, it is possible to improve a stability of the connection between the pad **10** and the solder. In particular, it is possible to significantly improve the durability of the solder against a force applied in the horizontal direction (the direction parallel to the upper surface **10a**) of the wiring board **2**.

(80) In addition, because the stepped surface **20s** is formed on the side surface of the groove **20g** of the insulating layer **20**, corner portions contacted by the solder flowing into the groove **20g** increases. Accordingly, because the force applied to the solder is dispersed, it is possible to improve the durability of the solder.

(81) <First Modification of Second Embodiment>

(82) A first modification of the second embodiment relates to an example of the wiring board having a pad with a different layer structure. In the first modification of the second embodiment, a description of the constituent elements that are the same as those of the embodiments described above may be omitted.

(83) FIG. **13A** and FIG. **13B** are diagrams illustrating an example of the wiring board according to the first modification of the second embodiment. FIG. **13A** is a partial plan view of the wiring board, and FIG. **13B** is a partial cross sectional view of the wiring board along a line D-D in FIG. **13A**.

(84) As illustrated in FIG. **13A** and FIG. **13B**, the wiring board **2A** differs from the wiring board **2** illustrated in FIG. **7A**, FIG. **7B**, or the like in that the pad **10** has a single-layer structure. The pad **10** is a copper layer, for example. The pad **10** includes the base portion **11b** and the extending portion **11s**, but does not include the metal layer **12**.

(85) The upper surface **10a** of the pad **10** (the upper surface of the base portion **11b**) is located at a position lower than the upper surface **20a** of the insulating layer **20**. The height of the upper surface **20a** of the insulating layer **20**, with reference to the upper surface of the base portion **11b** of the pad **10**, may be in a range of approximately 10 μm to approximately 70 μm , for example. Further, the upper surface **10a** of the pad **10** (the upper surface of the base portion **11b**) is located at a position lower than the stepped surface **20s**. The height of the stepped surface **20s**, with reference to the upper surface of the base portion **11b** of the pad **10**, may be in a range of approximately 2 μm to approximately 3 μm , for example. The upper surface of the extending portion **11s** is located at a position lower than the bottom surface of the groove **20g**. The height of the bottom surface of the groove **20g**, with reference to the upper surface of the extending portion **11s**, may be in a range of approximately 2 μm to approximately 3 μm , for example.

(86) An organic film that covers the upper surface and the side surface of the base portion **11b** of the pad **10**, and the upper surface of the extending portion **11s** of the pad **10**, may be provided, as required. The organic film includes an azole compound, an imidazole compound, or the like, for example. A thickness of the organic film may be approximately 1 μm or less, for example. Although the illustration of the organic film is omitted in FIG. **13B** to simplify the drawing, a coverage provided by the organic film will be the same as the coverage provided by the metal layer **12** in FIG. **12B**, for example.

(87) The wiring board **2A** illustrated in FIG. **13A** and FIG. **13B** can be manufactured by processes that are the same as those of the second embodiment described above in conjunction with FIG. **8A** through FIG. **12A**. After the process illustrated in FIG. **12A**, the organic film can be formed by performing an OSP process with respect to the base portion **11b** and the extending portion **11s**, as required.

(88) In a case where the pad **10** has the single-layer structure and the organic film is not famed, the etching process for removing the oxide film is not required.

(89) Accordingly to each of the embodiments and modifications described above, it is possible to provide a wiring board in which an outer edge of an external connection pad is disposed on the inner side of an outer edge of an opening in an insulating layer, and a method for manufacturing such a wiring board.

(90) Various aspects of the subject-matter described herein may be set out non-exhaustively in the following numbered clauses:

(91) 1. A method for manufacturing a wiring board, comprising: forming a first sacrificial layer on a support body, and further forming an annular second sacrificial layer protruding from an upper surface of the first sacrificial layer; forming a metal layer filling a recess formed by the first sacrificial layer and the second sacrificial layer, covering an inner periphery of an upper surface of the second sacrificial layer, and exposing an outer periphery of the upper surface of the second sacrificial layer; forming an insulating layer on the support body, so as to cover the first sacrificial layer, the second sacrificial layer, and the metal layer; and removing the support body, the first sacrificial layer, and the second sacrificial layer by etching, to form an external connection pad including the metal layer, wherein the first sacrificial layer and the second sacrificial layer are formed of a material that can be selectively etched with respect to the metal layer.

(92) 2. The method for manufacturing the wiring board according to clause 1, wherein the first sacrificial layer is formed in a predetermined region on the support body, and an outer edge of the second sacrificial layer is formed on an inner side of an outer edge of the first sacrificial layer in a plan view.

(93) Although the embodiments are numbered with, for example, “first,” or “second,” the ordinal numbers do not imply priorities of the embodiments. Many other variations and modifications will be apparent to those skilled in the art.

(94) All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiments of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A wiring board comprising: a pad configured to make an external electrical connection; and an insulating layer, wherein the pad includes a base portion, and an extending portion formed integrally with the base portion and extending toward an outer periphery of a side surface of the base portion in a plan view at a lower end of the side surface of the base portion, the insulating layer is provided with a groove that is located in a periphery of the pad in the plan view, the groove extends along and exposes the side surface of the base portion, and opens to an upper surface of the insulating layer, the extending portion is disposed at a bottom of the groove, and at least a portion of a lower surface of the extending portion, opposite from the base portion, is in contact with the insulating layer, and a side surface of the groove is located on an outer side of a side surface of the extending portion in the plan view.
2. The wiring board as claimed in claim 1, wherein an upper surface of the base portion is located at a position lower than the upper surface of the insulating layer.
3. The wiring board as claimed in claim 2, wherein a stepped surface is formed on the side surface of the groove of the insulating layer, and the upper surface of the base portion is located at a

position lower than the stepped surface.

4. The wiring board as claimed in claim 1, wherein an upper surface of the extending portion is located at a position lower than a bottom surface of the groove.

5. The wiring board as claimed in claim 1, further comprising: a metal layer covering an upper surface and the side surface of the base portion, and an upper surface of the extending portion.

6. The wiring board as claimed in claim 5, wherein an upper surface of the metal layer is located at a position higher than the upper surface of the insulating layer.

7. The wiring board as claimed in claim 5, wherein an upper surface of a portion of the metal layer laminated on the extending portion is located at a position higher than a bottom surface of the groove.

8. The wiring board as claimed in claim 1, further comprising: an organic film covering an upper surface and the side surface of the base portion, and an upper surface of the extending portion.

9. The wiring board as claimed in claim 2, wherein an upper surface of the extending portion is located at a position lower than a bottom surface of the groove.

10. The wiring board as claimed in claim 9, further comprising: a metal layer covering the upper surface and the side surface of the base portion, and the upper surface of the extending portion.

11. The wiring board as claimed in claim 1, further comprising: an interconnect layer formed on a lower surface of the insulating layer, wherein the interconnect layer includes a via interconnect penetrating the insulating layer and making contact with the lower surface of the extending portion.
